

Who Signed Up for the Do-Not-Call List?

Hal Varian
Fredrik Wallenberg
Glenn Woroch

Economics of the Information Economy
Universities Research Conference
National Bureau of Economic Research
Cambridge, MA
May 7 and 8, 2004

7 May 2004

Who Signed Up for the Do-Not-Call List?



Privacy as “solitude”

“The makers of our Constitution . . . sought to protect Americans in their beliefs, their thoughts, their emotions and their sensations. They conferred as against the Government, the right to be let alone—the most comprehensive of the rights of man and the right most valued by civilized men.”

—U.S. Supreme Court Justice Louis D. Brandeis.

- Increased number of portals open for solicitors
 - Fixed and mobile phones, fax machines
 - Postal mail, voice mail, email
 - Instant messaging, picture messaging
- Consumer options to protect their telephone privacy
 - Custom calling features (e.g., caller ID)
 - Voice mail, answering machines
 - Non-published, non-listed numbers
 - Do-not-call (“DNC”) registries

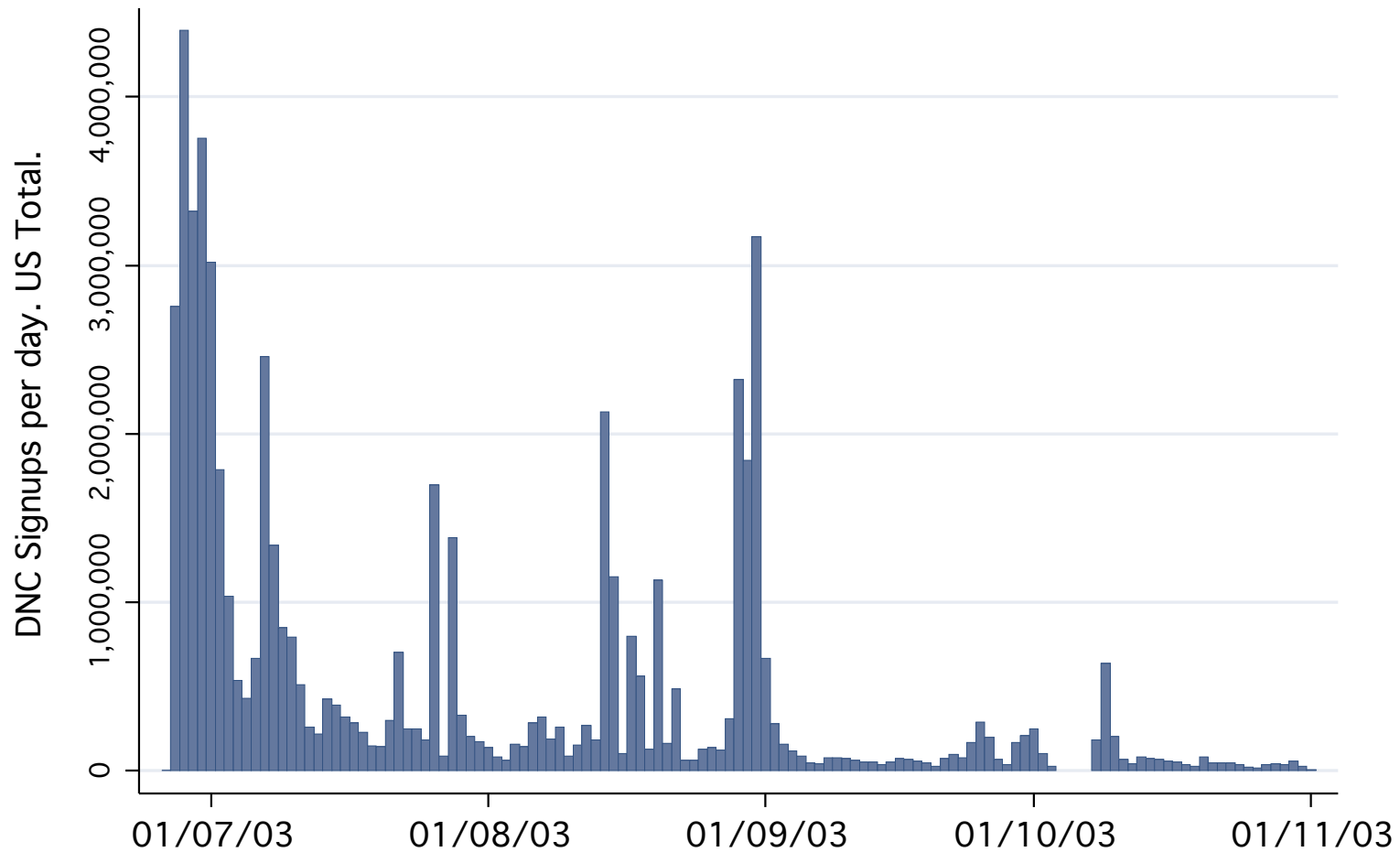


Do-not-call registries

- Principal registries
 - National and state do-not-call lists
 - Direct Marketing Association's "Telephone Preference Service" (TPS)
 - Company-specific lists
- Provisions of the national registry
 - Began June 27, 2003
 - Registration by phone and web
 - Both fixed and cellular, but only residential
 - No charge to consumer
 - Up to \$11,000 fines for violations

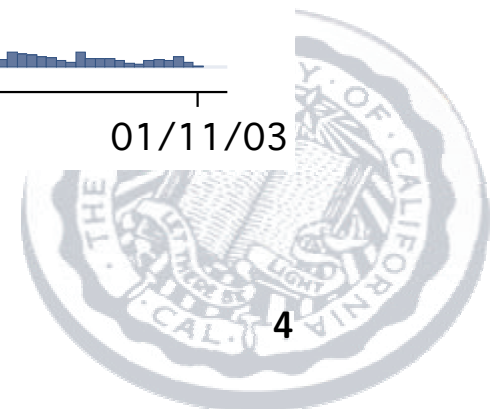


DNC sign-ups over time



7 May 2004

Who Signed Up for the Do-Not-Call List?



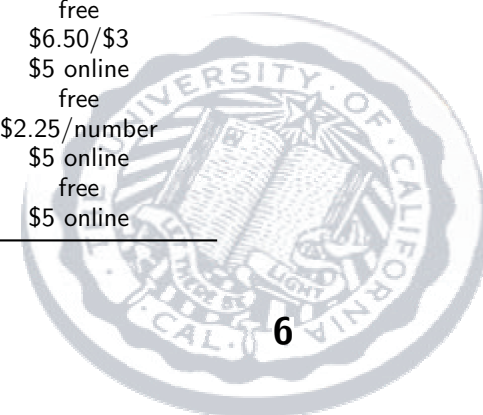
Interactions between national and state registries

- State programs
 - 31 states with registries (27 as of June 26, 2003)
 - 11 states charged residents for registration
 - 4 states relied on DMS's TPS
 - Some states mirror the provisions of the national list
- Coordination with national program
 - 15 states merged with national list
 - Other programs continue in parallel or shut down

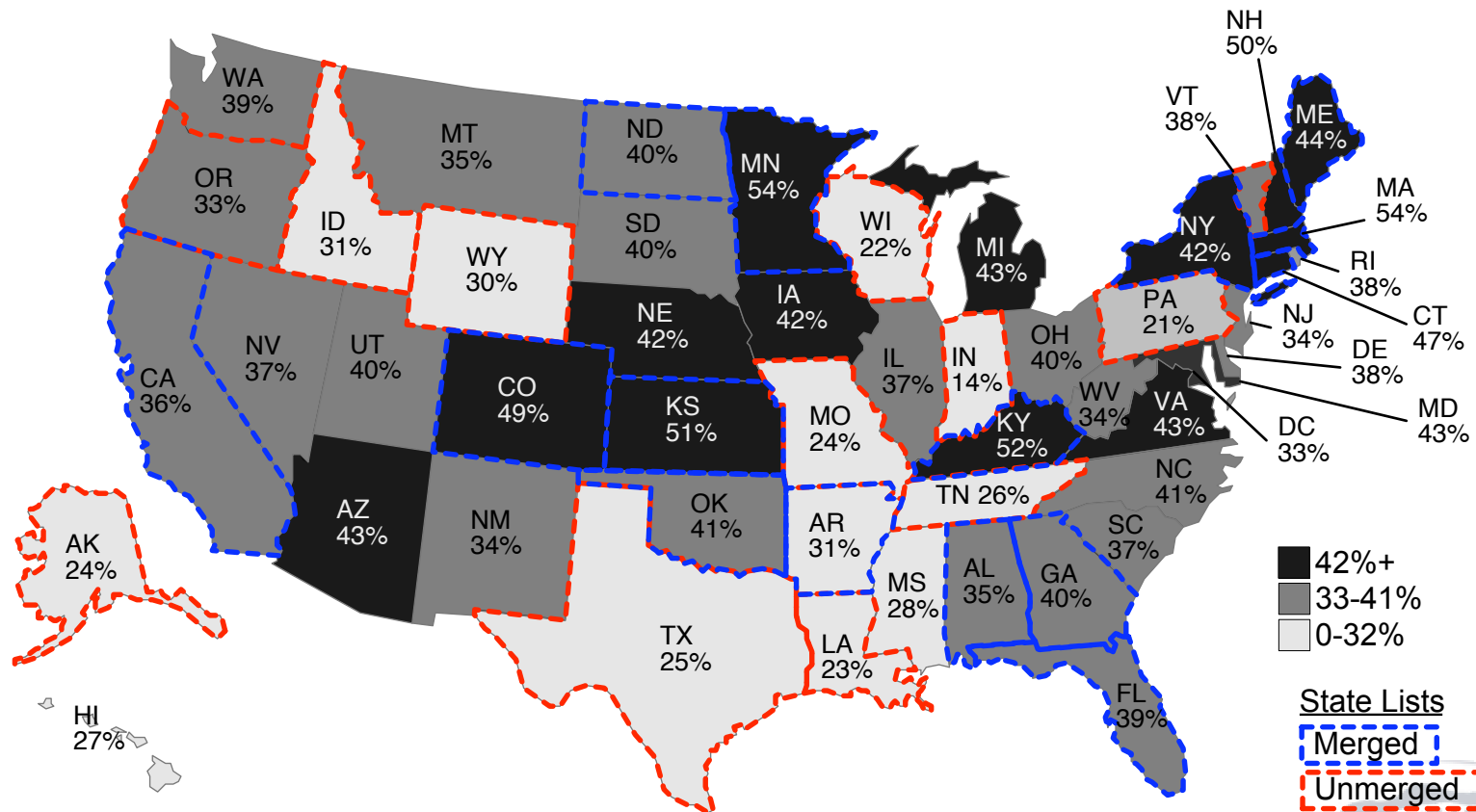


Interactions between national and state registries (cont'd)

State	Start	Merged	DMA/TPS	Fee
Alabama	6/29/00	20/8		free
Alaska	Nov'96			\$5-\$50
Arkansas	Jan'00	21/8		\$5/year
California	4/1/03	26/7		free
Colorado	7/1/02	28/7		free
Connecticut	1/1/01	22/8		free
Florida	Q2'99	9/8		\$10/\$5
Georgia	Jan'99			\$5
Georgia	Jan'99	8/10		\$5
Idaho	Jan'01			\$10/\$5
Indiana	1/1/02			free
Kansas	1/1/03	18/8		free
Kentucky	7/15/02	17/8		free
Louisiana	1/1/02			free
Maine	2003?	26/7		free
	Aug'01		Yes	\$5 online
Massachusetts	1/1/03	15/8		free
Minnesota	1/1/03	20/8		free
Missouri	7/1/01			free
New York	4/1/01	14/8		free
North Dakota	4/1/03?	9/8		free
Oklahoma	1/1/03	22/7		free
Oregon	Jan'00			\$6.50/\$3
Pennsylvania	4/2/02		Yes	\$5 online
Tennessee	7/1/00			free
Texas	1/1/02			\$2.25/number
Vermont	7/1/02		Yes	\$5 online
Wisconsin	1/1/03			free
Wyoming	Jul'01		Yes	\$5 online



DNC sign-ups per household by state



7 May 2004

Who Signed Up for the Do-Not-Call List?

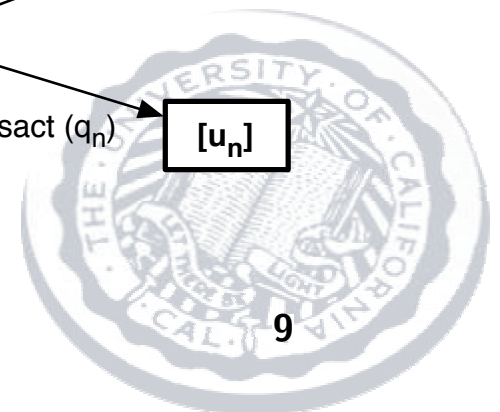
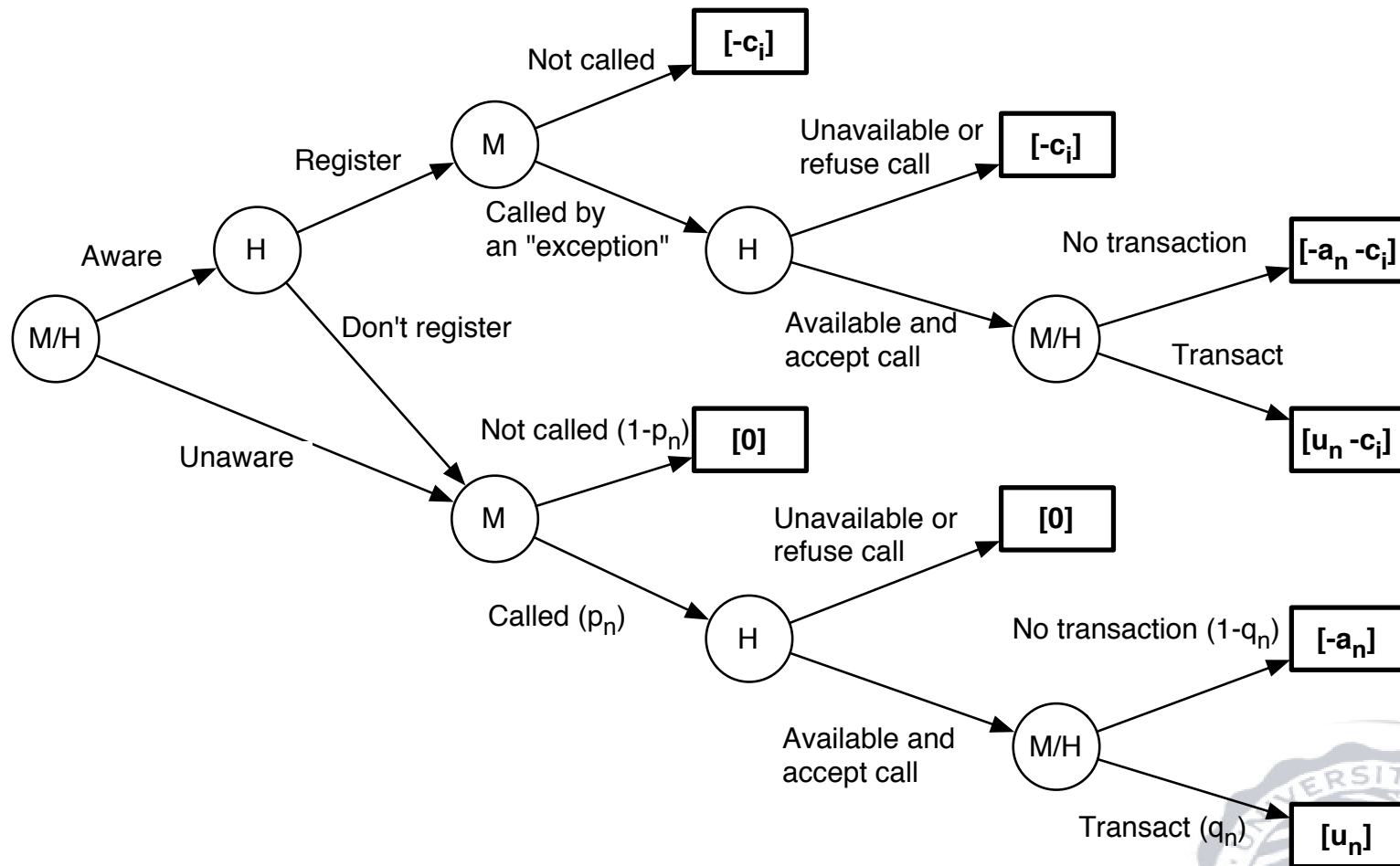
Research goals

In this paper we try to answer the following questions . . .

1. What factors correlates with people registering their phone numbers with the DNC list?
2. What monetary value does the average household attach to blocking telemarketing calls?
3. What impacts do the different registration modes (e.g., phone vs. web) have on sign-up frequencies?
4. What is the likelihood that consumers will register with a do-not-spam list?



Household decision to register on the DNC



Random utility model of household registration

Household utility:

$$u_{1n} = v_n(y_n - c_i, z_n) + \epsilon_{1n}$$

Probability of registration:

$$\phi_n = 1 - F[v_n(y_n, z_n) - v_n(y_n - c_i, z_n) - p_n a_n]$$

Some comparative statics under linear utility:

$$\frac{\partial \phi_n}{\partial c_i} = -f\beta_y < 0 \quad \text{and} \quad \frac{\partial \phi_n}{\partial a_n} = fp_n > 0$$

Average frequency of sign-up:

$$f_i = \frac{\sum_{n \in N_i} l_n \phi_n}{\sum_{n \in N_i} l_n}$$



Telemarketers' calling behavior

Expected profitability of a telemarketing call:

$$\pi_n = q_n(t_n - c) + (1 - q_n)(-c) + \nu_n = q_nt_n - c + \nu_n$$

- c = cost of a telemarketing call
- q_n = probability of successful transaction
- t_n = profitability of a successful transaction

Likelihood of calling a household:

$$p_n = Pr\{\pi_n > 0\} = Pr\{\nu_n > c - q_nt_n\} = 1 - G(c - q_nt_n)$$



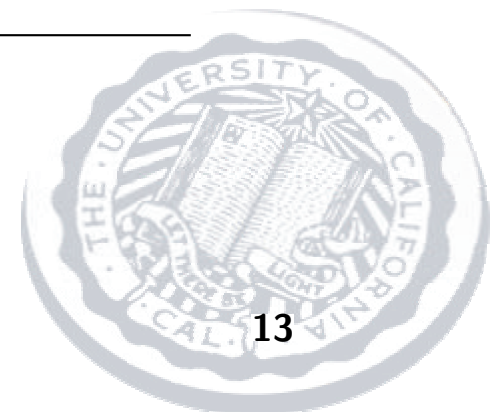
Data

- FTC do-not-call registry database
 - Redacted NPA-NXX phone numbers (“exchanges”)
 - All phone and web registrations during June 26 – Nov. 1, 2003
 - Includes time+date of sign-up
- Melissa database
 - Maps exchanges into 3,185 counties (5-digit FIPS)
 - Eliminate exchanges associated with cell phones
- 2000 Census data
 - Number of households and HHs with fixed line per county
 - CPS Supplement: survey of home use of Internet
- TNS Telecoms’ survey data
 - Quarterly survey of about 25,000 households, 3Q99 – 4Q01
 - Used to find average number of fixed lines per HH, Internet access



Summary statistics of key variables

Variable Name	Definition	Mean	Std. Dev.	Min.	Max.
pop	No. people	90611.83	294411.59	444	9519338
hh	No. households	33977.75	104956.02	185	3136279
Phone	No. fixed lines	33130.09	102818.27	178	3079273
dncland	Fixed line sign-ups	15389.64	49257.26	16	1311045
pDNC	Freq. of sign-ups	0.39	0.19	0.00	2.09
HRaceWhite	No. white HHs	26939.54	70845.25	122	1747061
HLatino	No. Latino HHs	2959.83	24450	0	1012351
HHInc_Med	Median HH income	35327.1	8826.86	15805	82929
HHPoverty	No. HHs below poverty line	3998.77	13998.70	22	474533
EduLow	Percent HHs with some HS or less	0.13	0.05	0.02	0.33
HHLingIso	No. HHs linguistically isolated	1407.37	11535.79	0	477729
OwnHome	No. HHs own home	22471.88	59832.62	118	1499694
HasMortgage	No. HHs have mortgage	12438.02	37999.45	8	1014178
HVal_Med	Avg. home value	84046.12	46198.72	20100	1000001
UnmarriedPartners	No. HHs with unmarried partners	1683.50	5686.34	0	181301
NoMale	No. HHs with no male	9866.75	33204.13	36	950073
pInternet	Percent HHs with Internet access	0.48	0.13	0.04	0.87
N		3094			



Demographics of sign-up frequencies

- Decompose HHs by demographic group
 - HH s in county i : $h_i = \sum_{g=1}^G n_{ig}$
 - Sign-ups in county i : $s_i = \alpha_1 n_{i1} + \dots + \alpha_G n_{iG}$
- Regression analysis of sign-up frequencies
 - Linear relationship between sign-up frequencies and demographic averages:

$$f_i = \frac{s_i}{h_i} = \alpha_1 \frac{n_{i1}}{h_i} + \dots + \alpha_G \frac{n_{iG}}{h_i}$$

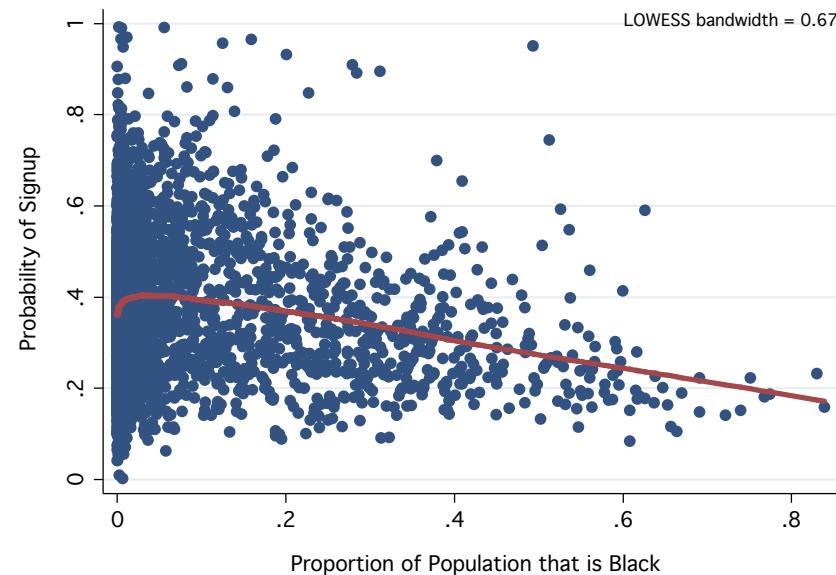
- Assume α_g are constant across counties
- Note: α_g is not a marginal effects since $\sum_{g=1}^G \alpha_g = 1$



Race regression

Variable	Coefficient	Std. Err.	Mean
pHHRace_White	0.396**	0.004	0.870
pHHRace_Black	0.155**	0.019	0.077
pHHRace_Native	−0.066	0.046	0.016
pHHRace_Asian	2.688**	0.218	0.006
pHHRace_PacIs	−14.072**	1.422	0.000
pHHRace_Other	−0.499**	0.079	0.018
pHHRace_Mult	2.125**	0.353	0.011

Significance levels : † : 10% * : 5% ** : 1%



7 May 2004

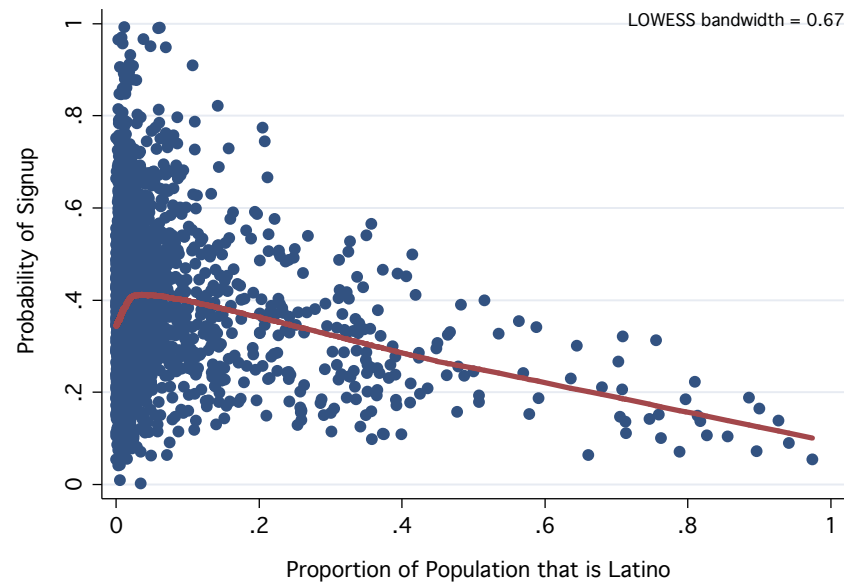
Who Signed Up for the Do-Not-Call List?



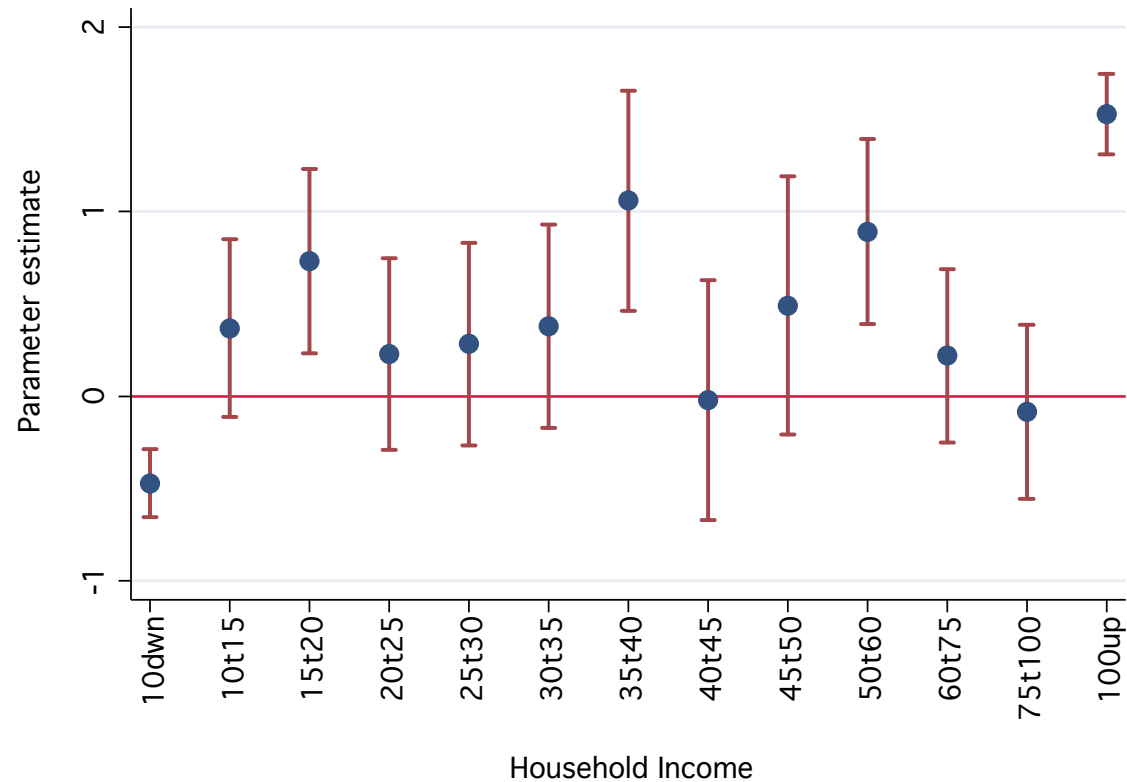
Latino regression

Variable	Coefficient	Std. Err.	Mean
pHHLatino	0.169**	0.026	0.045
pHHLatinoNo	0.392**	0.003	0.955
Difference	-0.223**	$F(1, 3092) = 66.28$	

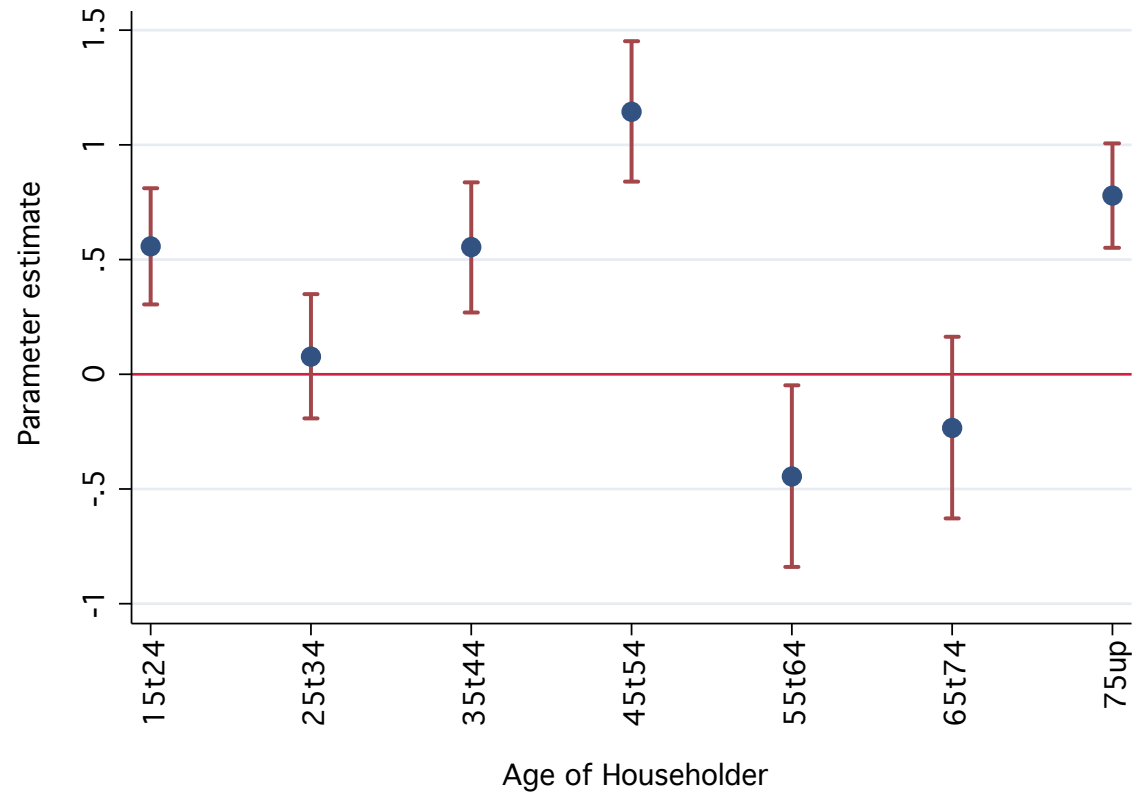
Significance levels : † : 10% * : 5% ** : 1%



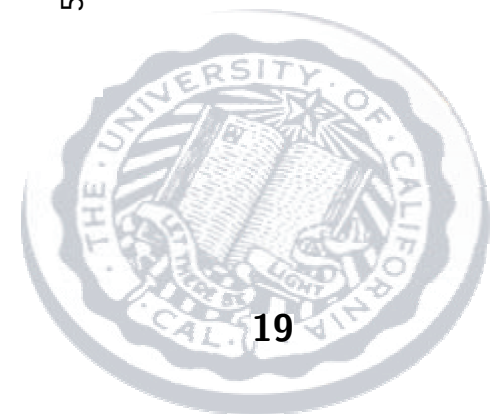
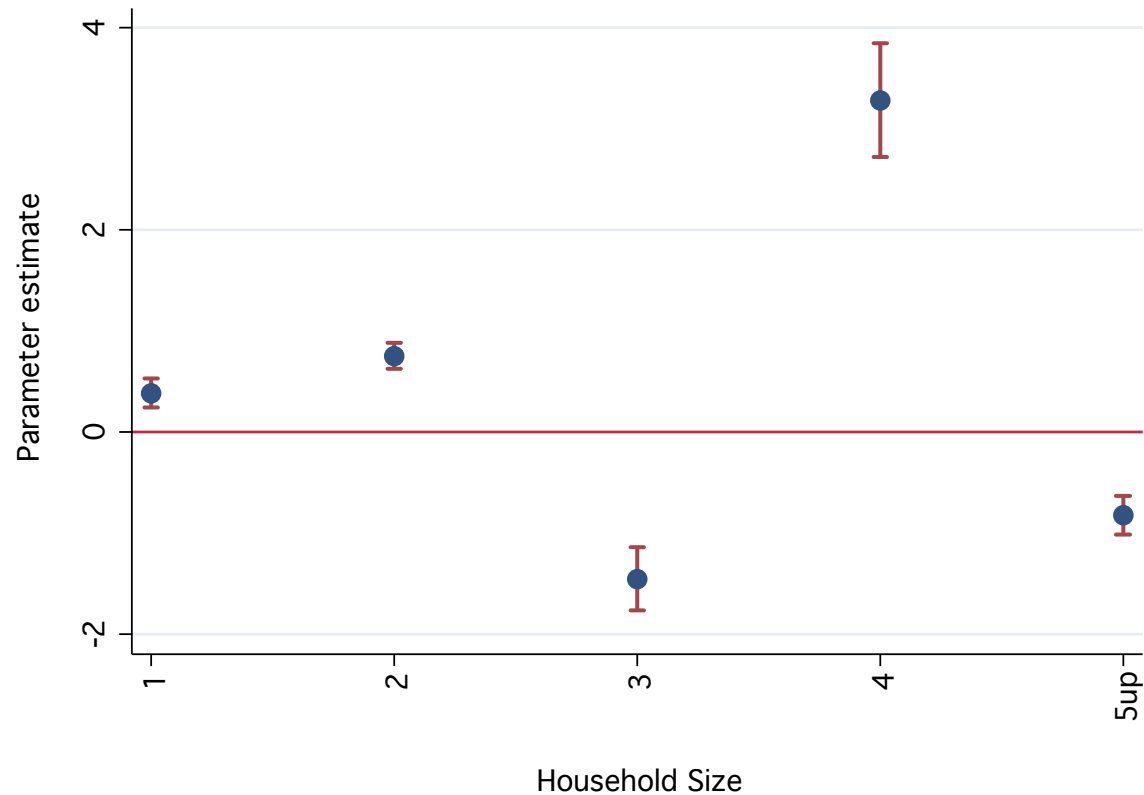
Household income regression



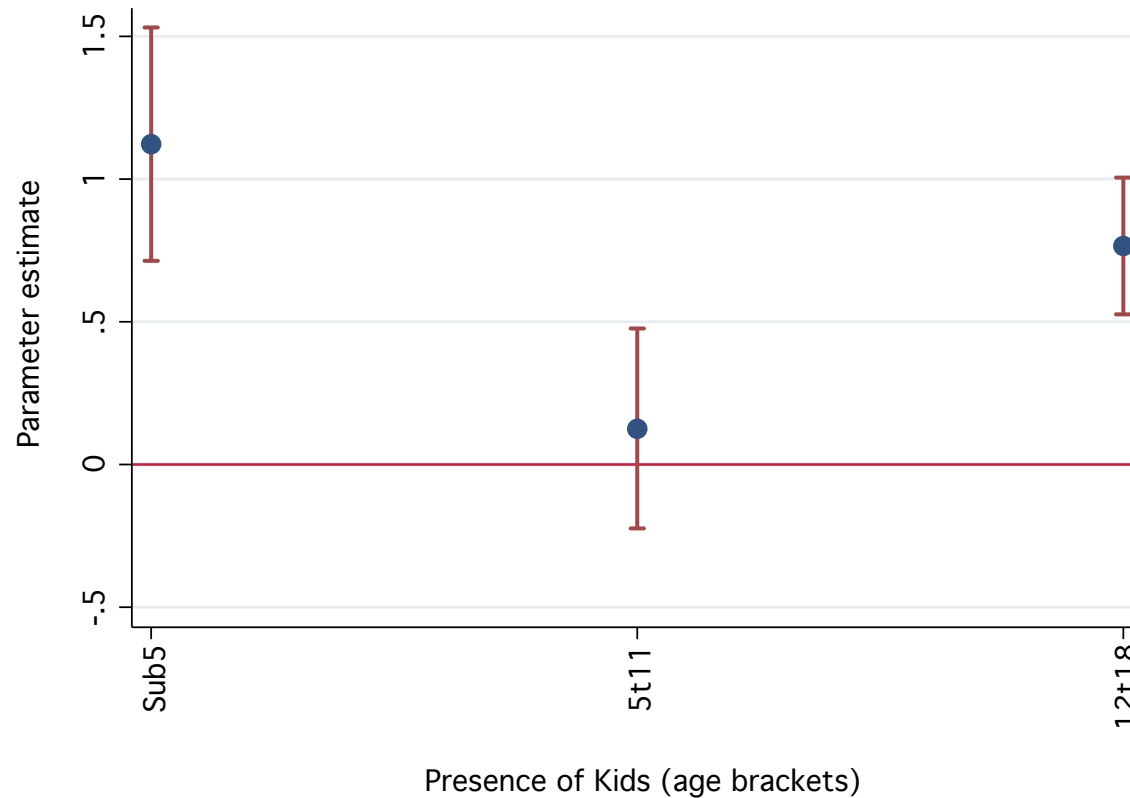
Householder age regressions



Household Size regressions



Presence of Kids regressions



Other demographic regressions

Demographic Group	Effect on Sign-up Frequency
Education	Low for some HS or less; high for college, postgrad
Linguistic isolation	Significantly lower
Home mortgage	Significantly higher
Poverty rate	Significantly lower
Unmarried partners	Significantly higher
Adult male present	Significantly higher
Internet usage at home	Somewhat higher
Urban v. rural	High in central city, but farm highest



Grouped logit models

- Logit specification of sign-up frequencies:

$$\log \left(\frac{f_i}{1 - f_i} \right) = \beta_0 + \beta_1 x_{1,i} + \cdots \beta_k x_{k,i} + \epsilon_i$$

- Estimate weighted regression with weights:

$$\sigma_i^2 = \frac{1}{n_i f_i (1 - f_i)}$$

- Eight specifications
 - Kitchen sink, Parsimonius I and II, State aggregation
 - With and without state dummies



Grouped logit results

- Goodness of fit
 - Small improvements in R^2 as throw in kitchen sink
 - State dummies contribute about 15%
- Coefficient estimates
 - Report estimates of log odds
 - Results of demographic regressions confirmed
 - State list which is merged (not merged) increases (decreases) sign-up



Grouped logit results

	Kitchen Sink		Parsimonius I		Parsimonius II	
	(1)	(2)	(3)	(4)	(5)	(6)
IHHInc: Med	4.561**	2.634**	2.746**	2.772**	2.017**	2.028**
pHHLatino	1.887**	2.017**	3.976**	2.759**		
pKids12t18	0.017**	0.050**	0.082**	0.084**	0.224**	0.258**
pHHLinglso	0.034**	0.045**	0.007**	0.011**		
p: EduLow	0.110**	0.079**	0.034**	0.035**	0.004**	0.002**
HasList	0.482**	0.773	0.499**	0.804	0.507**	0.674*
MergeList	2.568**	1.564*	2.459**	1.580*	2.344**	1.572*
Ksink Controls	Yes	Yes				
State Dummies		Yes		Yes		Yes
Adjusted R^2	0.61	0.75	0.58	0.72	0.55	0.70

Significance levels : † : 10% * : 5% ** : 1%; 3094 Observations



Income elasticities

	Model	Elasticity	Std. Err.	$P > z $
(1)	Kitchen Sink	0.939	0.090	0.000
(2)	Kitchen Sink, SD	0.604	0.089	0.000
(3)	Parsimonius I	0.626	0.029	0.000
(4)	Parsimonius I, SD	0.632	0.032	0.000
(5)	Parsimonius II	0.437	0.026	0.000
(6)	Parsimonius II, SD	0.441	0.028	0.000

Elasticities Calculated at the Median Income: \$35,348



Demand for do-not-spam

- Propensity to register for DNS similar to DNC

Spam	Telemarketing					
	DK	NA	None	Small	Big	Very big
DK	1	0	0	0	1	0
DNA	22	28	87	98	259	353
None	0	1	23	13	17	32
Small	0	0	7	64	69	65
Big	1	6	12	64	257	169
Very	1	5	23	36	152	334



Demand for do-not-spam (cont'd)

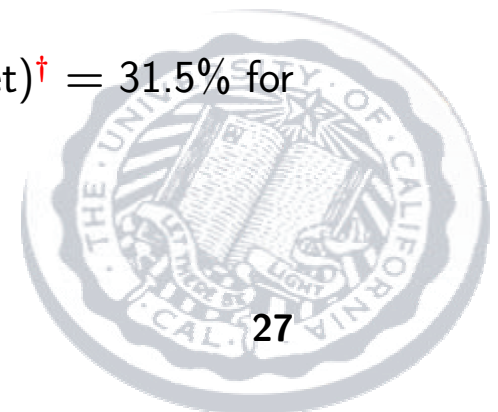
- Internet usage is nearly orthogonal to DNC sign-up
 - Correlation between using Internet access at home DNC sign-up is a mere 0.078.*
 - Natural experiment: no phone registration in east before July 7

	June 27–July 6			July 7–Nov 1
pInternet	1.026	1.026	0.656**	0.617**
east		1.345*	0.793	1.424*
east × pInternet			2.069**	1.215†
Adjusted R^2	0.79	0.79	0.79	0.80
Significance levels : † : 10% * : 5% ** : 1%				
Other controls: IHHInc_Med, pHHLatino, pKids12t18, pHHLingIso, pEduLow, HasList, MergeList and state dummies				

- Calculating sign-up rates for do-not-spam
 - (45% of HHs signed up for DNC) × (70% of HHs use Internet)[†] = 31.5% for DNS

*significant at the 0.01% level.

[†]From PEW's Internet Research.



Value of DNC registry

Two different back-of-the-envelope calculations ...

- Per call annoyance
 - 10 cents per call $\times$ 104 million calls per day = \$3.6 billion per year
- Willingness to pay from DMA TPS list
 - \$5 per year for 5 years $\Rightarrow$ \$1 per year
 - TPS reports 80% efficiency, assuming 100% efficiency for DNC $\Rightarrow$ \$1.25 per year.
 - 48 million registrants $\Rightarrow$ \$60 million per year



Extensions

- Feedback effects
 - Positive or negative feedback over time
 - * Negative externality registered HHs $\Rightarrow$ nonregistered
 - * Word of mouth effects of early popularity
 - * Time shifting of sign-ups?
 - Some indication that early sign-ups lead to increased later sign-ups
- Privacy calling features
 - Substitutes for DNC or complements?
 - Simple correlations tend to be small, but data aggregated
 - Perhaps substitution occurs at household level

